

EXAMPLE 14**Computation of Special Returns**

Let's return to Example 8. After reading this section, Mr. Lohrmann decided that he was not being fair to the fund manager by including the asset management fee and other expenses because the small size of the fund would put it at a competitive disadvantage. He learns that the fund spends a fixed amount of EUR500,000 every year on expenses that are unrelated to the manager's performance.

Mr. Lohrmann has become concerned that both taxes and inflation may reduce his return. Based on the current tax code, he expects to pay 20 percent tax on the return he earns from his investment. Historically, inflation has been around 2 percent and he expects the same rate of inflation to be maintained.

1. Estimate the annual gross return for the first year by adding back the fixed expenses.

Solution:

The gross return for the first year is higher by 1.67 percent ($= \text{EUR}500,000 / \text{EUR}30,000,000$) than the 15 percent investor return reported by the fund. Thus, the gross return for the first year is 16.67 percent ($= 15\% + 1.67\%$).

2. What is the net return that investors in the Rhein Valley Superior Fund earned during the five-year period?

Solution:

The investor return reported by the mutual fund is the net return of the fund after accounting for all direct and indirect expenses. The net return is also the pre-tax nominal return because it has not been adjusted for taxes or inflation. From Example 8, the net return for the five-year holding period was calculated as 42.35 percent.

3. What is the after-tax net return for the first year that investors earned from the Rhein Valley Superior Fund? Assume that all gains are realized at the end of the year and the taxes are paid immediately at that time.

Solution:

The net return earned by investors during the first year was 15 percent. Applying a 20 percent tax rate, the after-tax return that accrues to investors is 12 percent [$= 15\% - (0.20 \times 15\%)$].

4. What is the after-tax real return that investors would have earned in the fifth year?

Solution:

The after-tax return earned by investors in the fifth year is 2.4 percent [$= 3\% - (0.20 \times 3\%)$]. Inflation reduces the return by 2 percent so the after-tax real return earned by investors in the fifth year is 0.39 percent, as shown:

$$\frac{(1 + 2.40\%)}{(1 + 2.00\%)} - 1 = \frac{(1 + 0.0240)}{(1 + 0.0200)} - 1 = 1.0039 - 1 = 0.0039 = 0.39\%.$$

Note that taxes are paid before adjusting for inflation.